Appendix B

Symmetric Polynomials

Prerequisites for this chapter: Section A.1.

B.1 The Fundamental Theorem of Symmetric Polynomials

Given a commutative ring R (in our case we will consider $\mathbb{Z}$ and $\mathbb{Q}$) and an integer $n \geq 0$, we can consider the symmetric polynomials in n variables with coefficients in R. These are defined as the polynomials in n variables invariant under all permutations of these variables.

Definition B.1.1 (Symmetric Polynomials)

We say a polynomial $f \in R[X_1, \ldots, X_n]$ is *symmetric* if $f(X_1, \ldots, X_n) = f(X_{\sigma(1)}, \ldots, X_{\sigma(n)})$ for any permutation σ of $[n]$.

Exercise B.1.1. Let $f \in K(X_1, \ldots, X_n)$ be a rational function, where K is a field. Suppose f is symmetric, i.e. invariant under permutations of $X_1, \ldots, X_n$. Prove that $f = g/h$ for some symmetric polynomials $g, h \in K[X_1, \ldots, X_n]$.

As an example, $f = X^2Y + XY^2 + X^2 + Y^2$ is a symmetric polynomial in two variables, and

$$g = X^2YZ + XY^2Z + XYZ^2 + XY^2 + X^2Y + XZ^2 + X^2Z + YZ^2 + Y^2Z$$

is a symmetric polynomial in three variables.

Definition B.1.2 (Elementary Symmetric Polynomials)

The kth *elementary symmetric polynomial* for $k \geq 0$, $e_k \in R[X_1, \ldots, X_n]$, is defined by

$$e_k = \sum_{1 \leq i_1 < \ldots < i_k \leq n} X_{i_1} \cdot \ldots \cdot X_{i_k}.$$

Further, if $k > n$ then $e_k = 0$ (the empty sum) and if $k = 0$ then $e_0 = 1$ (the sum of the empty product).

The two-variable symmetric polynomials are thus simply $e_1 = X + Y$ and $e_2 = XY$. The three-variable ones are $e_1 = X + Y + Z$, $e_2 = XY + YZ + ZX$ and $e_3 = XYZ$.

We now state the fundamental theorem of symmetric polynomial.

Theorem B.1.1 (Fundamental Theorem of Symmetric Polynomials)

Suppose $f \in R[X_1, \ldots, X_n]$ is a symmetric polynomial. Then $f \in R[e_1, \ldots, e_n]$. In other words, there is a polynomial $g \in R[X_1, \ldots, X_n]$ such that

$$f(X_1, \ldots, X_n) = g(e_1, \ldots, e_n).$$

This theorem explains why we called e_k "elementary symmetric polynomials": because they generate all symmetric polynomials.

Proof

We proceed by induction on the number n of variables. For $n = 1$ this is trivial as $\sigma_1 = X$. Thus suppose it is true for $n - 1$ variables.

Now, we proceed again by induction on the degree d of f. When $f = 0$ this is obvious. Otherwise, by the first induction hypothesis,

$$f(X_1, \ldots, X_{n-1}, 0) = g(e'_1, \ldots, e'_{n-1})$$

for some $g \in K[X_1, \ldots, X_n]$, where e'_i is the ith elementary symmetric polynomial **in** $n - 1$ **variables**, i.e. $e'_i(X_1, \ldots, X_{n-1}) = e_i(X_1, \ldots, X_{n-1}, 0)$. Notice that $\deg g \leq \deg f$.

Write

$$f(X_1, \ldots, X_n) = g(e_1, \ldots, e_{n-1}) + h(X_1, \ldots, X_n).$$

We have $h(X_1, \ldots, X_{n-1}, 0) = 0$, i.e.

$$X_n \mid h(X_1, \ldots, X_{n-1}, X_n)$$

($h(X_1, \ldots, X_n)$ has a root at X_n in the ring $R[X_1, \ldots, X_{n-1}][X]$). By symmetry we also have $X_i \mid h$ for any i so $h = X_1 \cdot \ldots \cdot X_n r = e_n s$.

To conclude s is symmetric and has degree at most $\deg f - n < \deg f$ so $r = s(e_1, \ldots, e_n)$ by the second induction hypothesis and we get

$$f = g(e_1, \ldots, e_{n-1}) + e_n r(e_1, \ldots, e_n)$$

as wanted.

■

Exercise B.1.2. Prove that the decomposition of a symmetric polynomial f as $g(e_1, \ldots, e_n)$ is unique.

B.2 Newton's Formulas

Apart from elementary symmetric polynomials, there are another type of polynomials which are of particular interest.

Definition B.2.1 (Power Sum Polynomials)

The kth *power sum polynomial* for $k \geq 0$, $p_k \in R[X_1, \ldots, X_n]$, is defined by

$$p_k = X_1^k + \ldots + X_n^k.$$

Here is how they relate to the elementary symmetric polynomials.

Theorem B.2.1 (Newton's Formulas)

For any integer $k \geq 0$, we have

$$ke_k = \sum_{i=1}^{k} (-1)^{i-1} e_{k-i} p_i = e_{k-1}p_1 - e_{k-2}p_2 + \ldots + (-1)^{k-1} p_k.$$

The main importance of these formulas is that they let us recover the e_i from the p_i by induction, as the RHS only has e_i for $i < k$. This is expressed in the following corollary.

Corollary B.2.1*

Let K be a field of characteristic zero. Then, $K(e_1, \ldots, e_n) = K(p_1, \ldots, p_n)$. In particular, if

$$p_1(x_1, \ldots, x_n), \ldots, p_n(x_1, \ldots, x_n)$$

all lie in K, then so do

$$e_1(x_1, \ldots, x_n), \ldots, e_n(x_1, \ldots, x_n).$$

Remark B.2.1

This still holds in characteristic p as long as $n < p$, so that the k in ke_k is non-zero.

Exercise B.2.1*. Prove Corollary B.2.1.

Elementary Proof of Newton's Formulas

Let's compute the product $e_{k-1}p_i$. e_{k-i} is the sum of products of $k-i$ distinct variables, while p_i is the sum of the ith powers of all the variables. Thus, $e_{k-i}p_i$ is the sum of the product of the ith power of some variable, times $k-i$ distinct variables.

In the product of the $k-i$ distinct variable, either one of these variables will be the same as the one raised to the ith power, or not. Hence, we get $e_{k-i}p_i = r(i) + r(i+1)$ where $r(j)$ is the sum of products of one other variable raised to the $k-j$th power times j other distinct variables:

$$\sum_{i_1 < \ldots < i_{k-j}} X_{i_1} \cdot \ldots \cdot X_{i_{k-j}} \cdot \sum_{\ell \neq i_1, \ldots, i_{k-j}} X_\ell^j.$$

Thus,

$$\sum_{i=1}^{k} (-1)^{i-1} e_{k-i}p_i = (r(1)+r(2)) - (r(2)+r(3)) + \ldots + (-1)^k (r(k)+r(k+1)) = r(1) + (-1)^k r(k+1).$$

To conclude, $r(1) = ke_k$ and $r(k+1) = 0$ (there is no sum over -1 variables, or at least which is a homogeneous polynomial of degree k).

■

Proof of Newton's Formulas using Generating Functions

We now present a proof by generating functions. We work in $R[X_1, \ldots, X_k][[T]]$, i.e. the ring of

formal power series in T with coefficients in $R[X_1, \ldots, X_n]$. We have

$$(1 - X_1 T) \cdot \ldots \cdot (1 - X_n T) = \sum_{k=0}^{n} (-1)^k e_k T^k$$

so, by differentiating, we get

$$\begin{aligned}
\sum_{k=0}^{n} (-1)^k k e_k T^k &= T \sum_{i=1}^{n} \left((-X_i) \prod_{j \neq i} (1 - X_j T) \right) \\
&= - \left(\sum_{i=1}^{n} \frac{X_i T}{1 - X_i T} \right) \prod_{j=1}^{n} (1 - X_j T) \\
&= - \left(\sum_{i=1}^{n} \sum_{j=1}^{\infty} (X_i T)^j \right) \left(\sum_{i=0}^{n} (-1)^i e_i T^i \right) \\
&= \left(\sum_{i=1}^{\infty} p_i T^i \right) \left(\sum_{j=0}^{n} (-1)^{i-1} e_i T^i \right).
\end{aligned}$$

By comparing the T^k coefficients, we get

$$(-1)^k = \sum_{o=1}^{k} (-1)^{k-i-1} p_i e_{k-i}$$

as wanted.

■

B.3 The Fundamental Theorem of Algebra

Recall the statement of the fundamental theorem of algebra, whose name was coined at a time where algebra was about solving polynomial equations.[1] In reality, it is a theorem about analysis and is usually proven that way. However, in this section we will present a mostly algebraic one (a completely algebraic proof is impossible because $\mathbb{R}$ is defined analytically).

Theorem B.3.1 (Fundamental Theorem of Algebra)

Any polynomial $f \in \mathbb{C}[X]$ of degree $n \geq 0$ has exactly n roots, i.e.,

$$f = a(X - \alpha_1) \cdot \ldots (X - \alpha_n)$$

where $\alpha_1, \ldots, \alpha_n$ are the roots of f counted with multiplicity and a is its leading coefficient.

By Proposition A.1.2, it suffices to show that any non-constant polynomial $f \in \mathbb{C}[X]$ has a root in $\mathbb{C}$. This is also called the d'Alembert-Gauss theorem because d'Alembert was the first one to recognise the importance of proving this result but gave a flawed proof, while Gauss was (almost) the first one to give a rigorous proof (in fact he even gave multiple proofs).

[1] Now it means abstract algebra and linear algebra, see Section A.2 and Appendix C (the section on abstract algebra does not give justice to the subject, since it was only about setting up useful definitions for this book).

Theorem B.3.2 (d'Alembert-Gauss Theorem)

Any polynomial non-constant polynomial with complex coefficients has a complex root.

Notice that we can assume the polynomial f has real coefficients, since if f has complex coefficients, $g = f\overline{f}$ has real coefficients where $\overline{f}$ denotes the polynomial obtained by applying complex conjugation to the coefficients of f. Thus, if we find a root α of g, either $\alpha \in \mathbb{R}$ in which case we can use induction on $\frac{g}{X-\alpha}$, or we know by Proposition A.1.6 that $\overline{\alpha}$ is also a root and we can use induction on $\frac{g}{(X-\alpha)(X-\overline{\alpha})}$. This would prove that g has as many complex roots as its degree, and thus f too.

We shall only use two results. The first one is a corollary of the intermediate value theorem, while the second one is left as an exercise.

Proposition B.3.1

Any polynomial $f \in \mathbb{R}[X]$ of odd degree has a real root.

Proof

Since f has odd degree, $f(-\infty)$ and $f(+\infty)$ have opposite signs so there is a root by the intermediate value theorem. (Here we work in $\mathbb{R} \cup \{+\infty, -\infty\}$, which is just a nice shortcut for not writing limits.)

■

Proposition B.3.2

Any polynomial $f \in \mathbb{C}[X]$ of degree 2 has a root $\mathbb{C}$.

Exercise B.3.1*. Prove Proposition B.3.2.

Proof that any non-constant polynomial with real coefficients has a complex root

Let $f \in \mathbb{R}[X]$ be a polynomial of degree n. We proceed by induction on $k = v_2(n)$; the case $k = 0$ is Proposition B.3.1. For the induction step, suppose $k \geq 1$ so that n is even.

Let $\alpha_1, \ldots, \alpha_n$ be the roots of f. You might wonder what that means since we don't know if they exist (in $\mathbb{C}$) or not. It's true that we don't know whether they lie in $\mathbb{C}$ or not yet, but we can *construct them formally*, just like i was constructed formally to be an object such that $i^2 = -1$. Thus we can construct $\alpha_1, \ldots, \alpha_n$ inductively such that f has n roots in some field K. (However, if you add a formal object α such that $g(\alpha)$ to a field, this will make a field only if g is irreducible. But you can factorise f into a product of irreducible polynomials then add a root of one of the factors and repeat. See Exercise 4.2.1*.)

Given a real number t, we consider the polynomial

$$g_t = \prod_{i<j} X - (\alpha_i + \alpha_j + t\alpha_i\alpha_j)$$

which has real coefficients by the fundamental theorem of symmetric polynomials: if $\prod_{i<j} X - (X_i + X_j + tX_iX_j) = h_t(X, e_1, \ldots, e_n)$ for some $h_t \in \mathbb{R}[X][X_1, \ldots, X_n]$, then

$$g_t = h_t(X, e_1(\alpha_1, \ldots, \alpha_n), \ldots, e_n(\alpha_1, \ldots, \alpha_n))$$

since $e_i(\alpha_1, \ldots, \alpha_n) \in \mathbb{R}$ by Vieta's formulas.

Notice that g_t has degree $\frac{n(n-1)}{2}$ and, since n is even, $v_2(\deg g_t) = n-1$. Hence, g_t always has a complex root $\alpha = \alpha_i + \alpha_j + t\alpha_i\alpha_j$ for some i, j by the induction hypothesis.

Now pick $\frac{n(n-1)}{2} + 1$ values of $0 \neq t \in \mathbb{R}$: by the pigeonhole principle two of them must have the same indices, $\alpha_i + \alpha_j + r\alpha_i\alpha_j$ is a complex root of g_r and $\alpha_i + \alpha_j + s\alpha_i\alpha_j$ is a complex root of g_s for the same i, j.

By subtracting these two numbers, we get $\alpha_i\alpha_j \in \mathbb{C}$. Similarly, we have $\alpha_i + \alpha_j \in \mathbb{C}$. Thus α_i and α_j are roots of a quadratic equation $X^2 - (\alpha_i + \alpha_j)X + \alpha_i\alpha_j$ with complex coefficients and hence also lie in $\mathbb{C}$ by Proposition B.3.2. In particular, we have found a complex root of f as wanted.

■

B.4 Exercises

Newton's Formulas

Exercise B.4.1. Denote by $h_k \in R[X_1, \ldots, X_n]$ the kth *complete homogeneous polynomial*, i.e. the sum of all monomials of degree k. Prove that

$$kh_k = \sum_{i=1}^{k} h_{k-i}p_i = h_{k-1}p_1 + h_{k-2}p_2 + \ldots + p_k$$

for any $k \geq 0$.

Exercise B.4.2† (Hermite's Theorem)**.** Prove that a function $f : \mathbb{F}_p \to \mathbb{F}_p$ is a bijection if and only if $\sum_{a\in\mathbb{F}_p} f(a)^k$ is 0 for $k = 1, \ldots, p-2$ and -1 for $k = p-1$.

Exercise B.4.3†**.** Suppose that $\alpha_1, \ldots, \alpha_n$ are such that $\alpha_1^k + \ldots + \alpha_k^n$ is an algebraic integer for all n. Prove that $\alpha_1, \ldots, \alpha_k$ are algebraic integers.

Algebraic Geometry[2]

Exercise B.4.4† (Resultant)**.** Let R be a commutative ring, and $f, g \in R[X]$ be two polynomials of respective degrees m and n. For any integer $k \geq 0$, denote by $R_k[X]$ the subset of $R[X]$ consisting of polynomials of degree less than k. The *resultant* $\mathrm{Res}(f, g)$ is defined as the determinant of the linear map

$$(u, v) \mapsto uf + vg$$

from $R_m[X] \times R_n[X]$ to $R_{m+n}[X]$. Prove that, if $f = \sum_i a_i X^i$ and $g = \sum_i b_i X^i$, we have[3]

$$\mathrm{Res}(f,g) = \begin{vmatrix} a_0 & 0 & \cdots & 0 & b_0 & 0 & \cdots & 0 \\ a_1 & a_0 & \cdots & 0 & b_1 & b_0 & \cdots & 0 \\ a_2 & a_1 & \ddots & 0 & b_2 & b_1 & \ddots & 0 \\ \vdots & \vdots & \ddots & a_0 & \vdots & \vdots & \ddots & b_0 \\ a_m & a_{m-1} & \cdots & \vdots & b_n & b_{n-1} & \cdots & \vdots \\ 0 & a_m & \ddots & \vdots & 0 & b_n & \ddots & \vdots \\ \vdots & \vdots & \ddots & a_{m-1} & \vdots & \vdots & \ddots & b_{n-1} \\ 0 & 0 & \cdots & a_m & 0 & 0 & \cdots & b_n \end{vmatrix},$$

[2]The link with symmetric polynomials is quite feeble, I admit. I included this section because of the resultant, which is a polynomial symmetric in the roots of its arguments.

[3]This is an $(m+n) \times (m+n)$ matrix, with n times the element a_0 and m times the element b_0.

and, if $f = a\prod_i X - \alpha_i$ and $g = b\prod_j X - \beta_j$, then[4]

$$\mathrm{Res}(f,g) = a^m b^n \prod_{i,j} \alpha_i - \beta_j.$$

In addition, prove that $\mathrm{Res}(f,g) \in (fR[X] + gR[X])$.[5] Finally, prove that if $f, g \in \mathbb{Z}[X]$ are monic and $uf+vg = 1$ for some $u, v \in \mathbb{Z}[X]$, $\mathrm{Res}(f,g) = \pm 1$. (It is not necessarily true that $(fR[X]+gR[X])\cap R = \mathrm{Res}(f,g)R$ for specific polynomials f, g, but we always have $\mathrm{Res}(f,g) \in fR[X]+gR[X]$ by the previous point.)

Exercise B.4.5. Prove that

- $\mathrm{Res}(fg,h) = \mathrm{Res}(f,h)\,\mathrm{Res}(g,h)$ for any $f, gh \in R[X]$.
- $\mathrm{Res}(f-gh,g) = b^{k-n}\,\mathrm{Res}(f,g)$ for any $f, g, h \in R[X]$ where k is the degree of $f-gh$, n is the degree of g and b its leading coefficient.
- $\mathrm{Res}(F(f,g),G(f,g)) = \mathrm{Res}(F,G)^k\,\mathrm{Res}(f,g)^{mn}$ where $F, G, f, g \in R[X,Y]$ are homogeneous polynomials of respective degrees m, n, k, k. Here, by $\mathrm{Res}(A,B)$ for homogeneous $A, B \in R[X,Y]$, we mean $\mathrm{Res}(A(X,1),B(X,1))$.

Exercise B.4.6† (Hilbert's Nullstellensatz)**.** Let K be an algebraically closed field. Suppose that $f_1, \ldots, f_m \in K[X_1, \ldots, X_n]$ have no common zeros in K. Prove that there exist polynomials $g_1, \ldots, g_m$ such that

$$f_1 g_1 + \ldots + f_m g_m = 1.$$

Deduce that, more generally, if f is a polynomial which is zero at common roots of polynomials $f_1, \ldots, f_m$ (we do not assume anymore that they have no common roots), then there is an integer k and polynomials $g_1, \ldots, g_m$ such that

$$f^k = f_1 g_1 + \ldots + f_m g_m.$$

Exercise B.4.7† (Weak Bézout's Theorem)**.** Prove that two coprime polynomials $f, g \in K[X,Y]$ of respective degrees m and n have at most mn common roots in K. (Bézout's theorem states that they have exactly mn common roots counted with multiplicity, possibly at infinity.[6])

Exercise B.4.8†**.** Prove that $n+1$ polynomials $f_1, \ldots, f_{n+1} \in K[X_1, \ldots, X_n]$ in n variables are *algebraically dependent*, meaning that there is some non-zero polynomial $f \in K[X_1, \ldots, X_{n+1}]$ such that

$$f(f_1, \ldots, f_{n+1}) = 0.$$

Exercise B.4.9† (Transcendence Bases)**.** Let L/K be a field extension. Call a maximal set of K-algebraically independent elements of L a *transcendence basis*. Prove that, if L/K has a transcendence basis of cardinality n, then all transcendence bases have cardinality n. This n is called the *transcendence degree* $\mathrm{trdeg}_K L$. Finally, show that, if $L = K(\alpha_1, \ldots, \alpha_n)$ any maximal algebraically independent subset of $\alpha_1, \ldots, \alpha_n$ is a transcendence basis. (In particular $\mathrm{trdeg}_K L \leq n$.)

Exercise B.4.10†**.** Let K be an algebraically closed field which is contained in another field L. Suppose that $f_1, \ldots, f_m \in K[X_1, \ldots, X_n]$ are polynomials with a common root in L. Prove that they also have a common root in K.

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[4] In particular, the discriminant of f is $\frac{(-1)^{\frac{n(n-1)}{2}}}{a} \cdot \mathrm{Res}(f, f')$.

[5] In other words, the resultant provides an explicit value of a possible constant in Bézout's lemma for arbitrary rings (such as $\mathbb{Z}$).

[6] This requires some care: we need to define the multiplicity of common roots as well as what infinity means. See any introductory text to algebraic geometry, e.g. Sharevich [**shafarevich**]. See also the appendix on projective geometry of Silverman-Tate [26].

Miscellaneous

Exercise B.4.11† (ISL 2020 Generalised)**.** Let $n \geq 1$ be an integer. Find the maximal N for which there exists a monomial f of degree N which can not be written as a sum

$$\sum_{i=1}^{n} e_i f_i$$

with $f_i \in \mathbb{Z}[X_1, \ldots, X_n]$.

Exercise B.4.12† (Lagrange)**.** Given a rational function $f \in K[X_1, \ldots, X_n]$, we denote by G_f the set of permutations $\sigma \in \mathfrak{S}_n$ such that

$$f(X_1, \ldots, X_n) = f(X_{\sigma(1)}, \ldots, X_{\sigma(n)}).$$

Let $f, g \in K(X_1, \ldots, X_n)$ be two rational functions. If $G_f \subseteq G_g$, prove that there exists a rational function $r \in K[e_1, \ldots, e_n](X)$ such that

$$g = r \circ f.$$

Exercise B.4.13† (Iran Mathematical Olympiad 2012)**.** Prove that there exists a polynomial $f \in \mathbb{R}[X_0, \ldots, X_{n-1}]$ such that, for all $a_0, \ldots, a_{n-1} \in \mathbb{R}$,

$$f(a_0, \ldots, a_{n-1}) \geq 0$$

is equivalent to the polynomial $X^n + a_{n-1}X^{n-1} + \ldots + a_0$ having only real roots, if and only if $n \in \{1, 2, 3\}$.

Exercise B.4.14. Let $f \in K[X]$ be a monic polynomial with roots $\alpha_1, \ldots, \alpha_n$. Prove that its discriminant, as defined in Remark 1.3.2 or Exercise B.4.4† is equal to

$$(-1)^{\frac{n(n-1)}{2}} \cdot \prod_{i=1}^{n} f'(\alpha_i).$$

Deduce a formula for the discriminant of $X^n + aX + b$ and show that it is valid over any ring (not necessarily one where $X^n + aX + b$ has at most n roots[7]).

[7] Remember that Corollary A.1.1 is valid only over fields and integral domains.

Appendix B

Symmetric Polynomials

B.1 The Fundamental Theorem of Symmetric Polynomials

Exercise B.1.1. Let $f \in K(X_1, \ldots, X_n)$ be a rational function, where K is a field. Suppose f is symmetric, i.e. invariant under permutations of $X_1, \ldots, X_n$. Prove that $f = g/h$ for some symmetric polynomials $g, h \in K[X_1, \ldots, X_n]$.

Solution

Let $r = f/g$ be a symmetric rational function. We write it as

$$r = \frac{\prod_{\sigma \in \mathfrak{S}_n} f(\sigma(X_1, \ldots, X_n))}{g \prod_{\mathrm{id} \neq \sigma \in \mathfrak{S}_n} f(\sigma(X_1, \ldots, X_n))}.$$

The numerator is a symmetric polynomial so the denominator must be too since the quotient is. ■

Exercise B.1.2. Prove that the decomposition of a symmetric polynomial f as $g(e_1, \ldots, e_n)$ is unique.

Solution

This accounts to proving that $f(e_1, \ldots, e_n) = 0$ if and only if $f = 0$. This is clear by induction on n (trivial when $n = 1$). Let f be such a polynomial and suppose for the sake of a contradiction that $e_n \mid f$. If we set $X_n = 0$ we get

$$f(e_1, \ldots, e_{n-1}, 0) = 0$$

where the e_i are now the elementary symmetric polynomials in $X_1, \ldots, X_{n-1}$. By the induction hypothesis, this means that $f(X_1, \ldots, X_{n-1}, 0) = 0$, i.e. $X_n \mid f$. By symmetry, $e_n = X_1 \cdot \ldots \cdot X_n \mid f$, a contradiction. ■

B.2 Newton's Formulas

Exercise B.2.1*. Prove Corollary B.2.1.

Solution

We have $K(p_1, \ldots, p_n) \subseteq K(e_1, \ldots, e_n)$ by the fundamental theorem of symmetric polynomials, and the reverse inclusion comes from the Newton formulas by induction, as explained before. (We need the assumption that K is a field because the LHS of the Newton's formulas has a factor of k which we need to divide by in the inductive step, and we need K to have characteristic zero so that $k \neq 0$.) ■

B.3 The Fundamental Theorem of Algebra

Exercise B.3.1*. Prove Proposition B.3.2.

Solution

By the quadratic formula (or completing the square), solving quadratic equations is equivalent to finding square roots. Thus, let $a + bi \in \mathbb{C}$ be a complex number, with $a, b \in \mathbb{R}$. We wish to find a square root $x + iy$ or $a + bi$, i.e.

$$x^2 - y^2 + 2ixy = (x + iy)^2 = a + bi.$$

This means $x^2 - y^2 = a$ and $2xy = b$. This is equivalent to x^2 and $-y^2$ being roots of $X^2 - aX - b^2/4$ by Vieta's formulas. Since the constant coefficient is negative, the roots are real (e.g. by the intermediate value theorem), and since the product is negative, one is positive and one negative. Label the positive one as x^2 and the negative one as $-y^2$, take the square roots to find x and y and adjust the sign to have $2xy = b$. ■

B.4 Exercises

Newton's Formulas

Exercise B.4.2† (Hermite's Theorem)**.** Prove that a function $f : \mathbb{F}_p \to \mathbb{F}_p$ is a bijection if and only if $\sum_{a \in \mathbb{F}_p} f(a)^k$ is 0 for $k = 1, \ldots, p-2$ and -1 for $k = p-1$.

Solution

If f is a bijection, then this is Exercise A.3.11†. Now suppose that this condition holds. Newton's formulas (note $k \neq 0$ for $k < p$ so Corollary B.2.1 still holds) tell us that there is only one possible value of $e_k(f(0), \ldots, f(p-1))$ for any fixed k. Hence, we must have

$$e_k(f(0), \ldots, f(p-1)) = e_k(0, \ldots, p-1)$$

since $0, \ldots, p-1$ satisfy the condition. This implies that

$$(X - f(0)) \cdot \ldots \cdot (X - f(p-1)) = (X - 0) \cdot \ldots \cdot (X - (p-1))$$

so f is a bijection as wanted. ■

Exercise B.4.3†. Suppose that $\alpha_1, \ldots, \alpha_n$ are such that $\alpha_1^k + \ldots + \alpha_k^n$ is an algebraic integer for all n. Prove that $\alpha_1, \ldots, \alpha_k$ are algebraic integers.

Solution

Newton's formulas give us $k!e_i(\alpha_1,\dots,\alpha_k) \in \overline{\mathbb{Z}}$ for any i. Thus, $k!\alpha \in \overline{\mathbb{Z}}$ for any $\alpha = \alpha_i$, by Exercise 1.5.22†. In particular, since the statement is also true when we replace the α_i by α_i^m for any fixed m, we get $k!\alpha^m \in \overline{\mathbb{Z}}$ for any m.

Thus, the problem reduces to showing that, if $\alpha \in \overline{\mathbb{Q}}$ is algebraic and such that $N\alpha^n \in \overline{\mathbb{Z}}$ (i.e. powers of α have bounded denominator) for some non-zero $N \in \mathbb{Z}$ and any positive integer n, then $\alpha \in \overline{\mathbb{Z}}$. For large n, the degree of α^{2^n} is constant, since the sequence

$$[\mathbb{Q}(\alpha^{2^n}) : \mathbb{Q}] = [\mathbb{Q}(\alpha^{2^n}) : \mathbb{Q}(\alpha^{2^{n-1}})][\mathbb{Q}(\alpha^{2^{n-1}}) : \mathbb{Q}]$$

is a non-increasing sequence of integers. By replacing α by α^{2^m} for some large m, we may assume that this is true for any $n \geq 0$. Let $\beta_1,\dots,\beta_\ell$ be the conjugates of α. Consider the minimal polynomial

$$f_{2^k} = \prod_{i=1}^{\ell} X - \beta_i^{2^k}$$

of α^{2^k} and let $N_k = 1/c(f_{2^k}))$ be the smallest positive integer such that $N_k f_{2^k} \in \mathbb{Z}[X]$. By assumption N_k is bounded. However, we have

$$\begin{aligned} N_k^2 f_{2^{k+1}}(X^2) &= N_k^2 \prod_{i=1}^{\ell} X^2 - \beta_i^{2^{n+1}} \\ &= \left(N_k \prod_{i=1}^{\ell} X - \beta_i^{2^n}\right)\left(N_k \prod_{i=1}^{\ell} X + \beta_i^{2^n}\right) \\ &= \pm(N_k f_{2^k})(N_k f_{2^k}(-X)) \end{aligned}$$

which is primitive by Gauss' lemma 5.1.2. Hence, $N_{k+1} = N_k^2$ so N_1 must be 1 otherwise $N_k = N_1^{2^{n-1}} \to \infty$. This means that the minimal polynomial of α has integral coefficients, i.e. α is an algebraic integer. ■

Remark B.4.1

It is necessary to mention that the key claim admits a very short and intuitive proof if we allows ourself some ideal theory. The idea is that, if $\alpha \in \mathbb{Q}$, we can simply look at the p-adic valuations to get $nv_p(\alpha) + v_p(N) \geq 0$ which gives us $v_p(\alpha) \geq 0$ for large enough n. Hence, α is an integer. For arbitrary algebraic integers, the same proof works almost verbatim: a number field K is not always a UFD but always has ideal factorisation. This means that we can this time consider prime ideals $\mathfrak{p}$ of K to get $nv_{\mathfrak{p}}(\alpha) + v_{\mathfrak{p}}(N) \geq 0$ which implies $v_{\mathfrak{p}}(\alpha) \geq 0$ again. Finally, since this is true for any prime ideal $\mathfrak{p}$, we get $\alpha \in \mathcal{O}_K$.

Algebraic Geometry

Exercise B.4.4† (Resultant)**.** Let R be a commutative ring, and $f, g \in R[X]$ be two polynomials of respective degrees m and n. For any integer $k \geq 0$, denote by $R_k[X]$ the subset of $R[X]$ consisting of polynomials of degree less than k. The *resultant* $\mathrm{Res}(f, g)$ is defined as the determinant of the linear map

$$(u, v) \mapsto uf + vg$$

from $R_m[X] \times R_n[X]$ to $R_{m+n}[X]$. Prove that, if $f = \sum_i a_i X^i$ and $g = \sum_i b_i X^i$, we have[1]

$$\mathrm{Res}(f,g) = \begin{vmatrix} a_0 & 0 & \cdots & 0 & b_0 & 0 & \cdots & 0 \\ a_1 & a_0 & \cdots & 0 & b_1 & b_0 & \cdots & 0 \\ a_2 & a_1 & \ddots & 0 & b_2 & b_1 & \ddots & 0 \\ \vdots & \vdots & \ddots & a_0 & \vdots & \vdots & \ddots & b_0 \\ a_m & a_{m-1} & \cdots & \vdots & b_n & b_{n-1} & \cdots & \vdots \\ 0 & a_m & \ddots & \vdots & 0 & b_n & \ddots & \vdots \\ \vdots & \vdots & \ddots & a_{m-1} & \vdots & \vdots & \ddots & b_{n-1} \\ 0 & 0 & \cdots & a_m & 0 & 0 & \cdots & b_n \end{vmatrix},$$

and, if $f = a\prod_i X - \alpha_i$ and $g = b\prod_j X - \beta_j$, then[2]

$$\mathrm{Res}(f,g) = a^m b^n \prod_{i,j} \alpha_i - \beta_j.$$

In addition, prove that $\mathrm{Res}(f,g) \in (fR[X] + gR[X])$.[3] Finally, prove that if $f, g \in \mathbb{Z}[X]$ are monic and $uf+vg = 1$ for some $u, v \in \mathbb{Z}[X]$, $\mathrm{Res}(f,g) = \pm 1$. (It is not necessarily true that $(fR[X]+gR[X]) \cap R = \mathrm{Res}(f,g)R$ for specific polynomials f, g, but we always have $\mathrm{Res}(f,g) \in fR[X]+gR[X]$ by the previous point.)

Solution

The determinant form of the resultant simply follows from considering the matrix of the linear function corresponding to the basis $1, X, \ldots, X^{m+n-1}$. To prove the explicit formula, consider the case where $A = a$, $B = b$, $\alpha_i = A_i$ and $\beta_j = B_j$ are indeterminates. Working over a field K, the resultant vanishes when $A_i = B_j$ for some i, j since the map is not surjective: it never reaches 1. Thus, the resultant is divisible by $A_i - B_j$ for all i, j. Looking at the determinant formula, we see that the degree of $\mathrm{Res}(f,g)$ in A_1 is n and its leading coefficient is $A^m B^n$, which proves the wanted formula.

For the second part, write the equation $uf+vg = r$ in the monomial basis as $RV = (r, 0, \ldots, 0) := re_1$, where R is the matrix corresponding to the linear map $(u, v) \mapsto uf + vf$. Hence, we wish to have $rR^{-1}e_1 \in R^n$. **??** tells us that $r = \det R = \mathrm{Res}(f,g)$ works.

Now let f and g be generic polynomials with integer coefficients of respective degree m and n.

Suppose finally that $(f\mathbb{Z}[X] + g\mathbb{Z}[X]) \cap \mathbb{Z} = \mathbb{Z}$. Write f and g as $\prod_{i=1}^m X - \alpha_i$ and $\prod_{j=1}^n X - \beta_i$. We have $u(\beta_i)f(\beta_i) = 1$ for each i, so

$$\prod_{i=1}^{f}(\beta_i) = \pm\,\mathrm{Res}(f,g)$$

divides 1 as desired. ■

Exercise B.4.6† (Hilbert's Nullstellensatz)**.** Let K be an algebraically closed field. Suppose that $f_1, \ldots, f_m \in K[X_1, \ldots, X_n]$ have no common zeros in K. Prove that there exist polynomials $g_1, \ldots, g_m$ such that

$$f_1 g_1 + \ldots + f_m g_m = 1.$$

[1] This is an $(m+n) \times (m+n)$ matrix, with n times the element a_0 and m times the element b_0.

[2] In particular, the discriminant of f is $\frac{(-1)^{\frac{n(n-1)}{2}}}{a} \cdot \mathrm{Res}(f, f')$.

[3] In other words, the resultant provides an explicit value of a possible constant in Bézout's lemma for arbitrary rings (such as $\mathbb{Z}$).

Deduce that, more generally, if f is a polynomial which is zero at common roots of polynomials $f_1, \ldots, f_m$ (we do not assume anymore that they have no common roots), then there is an integer k and polynomials $g_1, \ldots, g_m$ such that

$$f^k = f_1 g_1 + \ldots + f_m g_m.$$

Solution

We proceed by induction on the number n of variables. When $n = 1$ this is just Bézout's lemma. Now, if $n \geq 1$, we will eliminate one variable with the resultant. Consider the polynomial

$$g = \mathrm{Res}_{X_n}(f_m, U_1 f_1 + \ldots + U_{m-1} f_{m-1}) \in K[U_1, \ldots, U_{m-1}][X_1, \ldots, X_{n-1}],$$

where $U_1, \ldots, U_{m-1}$ are formal variables. By Exercise B.4.4†, $(x_1, \ldots, x_{n-1})$ is a root of g if and only if f_m and $U_1 f_1 + \ldots + U_{m-1} f_{m-1}$ have a common root x_n at $(x_1, \ldots, x_{n-1})$, i.e. $(x_1, \ldots, x_n)$ is a common root of $f_1, \ldots, f_m$, or if the leading coefficient in X_n of f_m and $U_1 f_1 + \ldots + U_{m-1} f_{m-1}$ vanish at $(x_1, \ldots, x_{n-1})$, i.e. the leading coefficient in X_n of $f_1, \ldots, f_m$ vanish at $(x_1, \ldots, x_{n-1})$ (we say $(x_1, \ldots, x_{n-1})$ is a common root at infinity). We wish to rule out the second case. This is not very hard: perform the change of coordinates $X_i \to X_i + c_i X_n$ for $i = 1, \ldots, n-1$ and some c_i to get constant leading coefficients in X_n (thus sharing no common root).

Hence, g has no root by assumption since $f_1, \ldots, f_m$ have no common root. However, a root of g is simply a common root of its coefficients g_i when g is seen as a polynomial in $U_1, \ldots, U_{m-1}$. This implies that a linear combination of the g_i is 1, by the induction hypothesis. Finally, notice that

$$g = \mathrm{Res}_{X_m}(f, U_1 f_1 + \ldots + U_{m-1} f_{m-1}) = uf + v(U_1 f_1 + \ldots + U_{m-1} f_{m-1})$$

for some $u, v \in K[X_1, \ldots, X_n][U_1, \ldots, U_{m-1}]$, by Exercise B.4.4†. Hence, the coefficients g_i of g are linear combinations of the f_i (with coefficients in $K[X_1, \ldots, X_n]$). We conclude that a linear combination of the f_i is 1 as wanted.

For the second part, suppose without loss of generality that $f \neq 0$. Use the first part on $f_1, \ldots, f_m, 1 - X_{n+1} f$ which have no common root by assumption (this is known as Rabinowitsch's trick). Thus, there are $g_1, \ldots, g_m, g \in K[X_1, \ldots, X_{n+1}]$ such that

$$g_1 f_1 + \ldots + g_m f_m + g(1 - X_{n+1} f) = 1.$$

Now, evaluate this at $X_{n+1} = 1/f$ and multiply by a large enough power of f to get the wanted equality. ■

Exercise B.4.7† (Weak Bézout's Theorem)**.** Prove that two coprime polynomials $f, g \in K[X, Y]$ of respective degrees m and n have at most mn common roots in K. (Bézout's theorem states that they have exactly mn common roots counted with multiplicity, possibly at infinity.[4])

Solution

We can assume without loss of generality that K has as many elements as we want by iteratively adding new elements to K using Exercise 4.2.1*.)

We shall proceed as in Exercise B.4.6†. Consider the resultant $h = \mathrm{Res}_Y(f, g)$. This is a polynomial of degree at most mn by its matrix expression of Exercise B.4.4†. By the same exercise, if (x, y) is a common root of f, g, then x is a root of h. Thus, we would be done if there was at most one possible value of y for each x, since h has degree at most mn and thus

[4] This requires some care: we need to define the multiplicity of common roots as well as what infinity means. See any introductory text to algebraic geometry, e.g. Sharevich [**shafarevich**]. See also the appendix on projective geometry of Silverman-Tate [26].

has at most mn roots. Note that we already get that there are finitely many common roots (although that's already a consequence of Bézout's lemma). Here is how we can achieve that: do a change of coordinates $X \to X + c'Y$ for some c chosen so that each x appears at most once as a common root (x, y) of f and g: this is possible because the common roots in this new system of coordinates are $(\alpha + c'\beta, \beta)$ and there are finitely many c' for which

$$\alpha + c'\beta = \alpha' + c'\beta' \iff c = \frac{\alpha - \alpha'}{\beta' - \beta}.$$

■

Exercise B.4.8†. Prove that $n+1$ polynomials $f_1, \dots, f_{n+1} \in K[X_1, \dots, X_n]$ in n variables are *algebraically dependent*, meaning that there is some non-zero polynomial $f \in K[X_1, \dots, X_{n+1}]$ such that

$$f(f_1, \dots, f_{n+1}) = 0.$$

Solution

We present two solutions: one with linear algebra and one with resultants.

For the first solution, consider the linear system of equations in $(N+1)^{n+1}$ variables

$$\sum_{i_1, \dots, i_{n+1} \leq N} a_{i_1, \dots, i_{n+1}} f_1^{i_1} \cdot \ldots \cdot f_{n+1}^{i_{n+1}} = 0. \tag{$*$}$$

We wish to find a non-trivial solution to this system. Let us count the number of equations we have. Set $M = \max_i(\deg f_i)$. Then, the LHS of $(*)$ is a polynomial of degree $(n+1)MN$, when we consider the $a_{i_1, \dots, i_{n+1}}$ as formal variables. Hence, we have $(N+1)^{n+1}$ unknowns and

$$\sum_{k=0}^{N} ((n+1)MN)^k = \frac{((n+1)MN)^{n+1} - 1}{(n+1)MN - 1}$$

equations, one for each coefficient. For large N, $(N+1)^{n+1} > \frac{((n+1)MN)^{n+1}-1}{(n+1)MN-1}$, which means that there is a non-trivial solution as wanted (the kernel is non-trivial by e.g. the rank-nullity theorem **??**, or Proposition C.1.2).

To make the idea of the second solution clearer, we treat the case $n = 1$ first. If $f, g \in K[X]$ are polynomials, the resultant $h = \mathrm{Res}_X(f - S, g - T)$ is a non-zero polynomial in S, T with coefficients in K. Indeed, it is non-zero since when $S - f$ and $T - g$ are coprime it takes a non-zero value (we can choose $T = 0$ and $S \in K$ to be a large constant for instance). However, when $S = f$ and $T = g$, the polynomials $f - S$ and $g - T$ are not coprime anymore so $h(f, g) = 0$ as wanted.

Now, we construct by backwards induction on k a polynomial with coefficients in $K[X_1, \dots, X_k]$ vanishing at $f_1, \dots, f_{n+1}$. In other words, we eliminate one variable each time. Here is how we do it: at first, $f_{n,i} = f_i$. Then, we define the polynomials

$$f_{k-1,i} = \mathrm{Res}_{X_k}(f_{k,k+1} - T_{k,k+1}, f_{k,i} - T_{k,i})$$

for $i = 1, \dots, k$. At each step we get rid of X_k and introduce $k+1$ new variables. Thus, $f_{0,1} \in K[\{T_{i,j} \mid i \leq j - 1\}$. It is clear that it is zero when evaluated at $T_{n,i} = f_i$ for every i and $T_{k,i}$ constant for $i \leq k - 1 \leq n - 2$. Indeed, note that $\mathrm{Res}(A, B)(t)$ is not in general equal to $\mathrm{Res}(A(t), B(t))$, since $A(t), B(t)$ do not have the same degree as A, B. If we consider constant polynomials as polynomials of degree $\deg(f_{k,i} - T_{k,i}) > 0$, then

$$\mathrm{Res}_{X_k}(f_{k,k+1} - T_{k,k+1}, f_{k,i} - T_{k,i}) = 0,$$

as can be seen from the matrix expression of Exercise B.4.4†. It remains to prove that there is some choice of such $T_{k,i}$ for which $f_{0,1}$ is not the zero polynomial. This is easy to see: we can choose $T_{k,k+1} = 0$ for all k and at each step we pick $T_{k,i}$ so that $f_{k,k+1}$ and $f_{k,i} + T_{k,i}$ are coprime. Indeed, if $f_{k,k+1}$ has ℓ irreducible prime factors, if we pick $\ell + 1$ values of $T_{k,i}$ one of them must work, as otherwise we would have

$$\pi \mid (f_{k,i} + T_{k,i}) - (f_{k,i} - T'_{k,i}) = T_{k,i} - T'_{k,i}$$

for some irreducible $\pi \mid f_{k,k+1}$ and distinct $T_{k,i}, T'_{k,i} \in K$ by the pigeonhole principle. This is impossible sine it implies $T_{k,i} = T'_{k,i}$. There is still one slight technicality: we could have $\ell \geq |K|$. However, we can simply add elements to K to get a sufficiently large K as in Exercise B.4.7†, and then consider the norm of the polynomial f we obtain (i.e. take the product over each of its conjugates, exactly like we did in the solution of Exercise 1.5.22†). ■

Exercise B.4.9† (Transcendence Bases)**.** Let L/K be a field extension. Call a maximal set of K-algebraically independent elements of L a *transcendence basis*. Prove that, if L/K has a transcendence basis of cardinality n, then all transcendence bases have cardinality n. This n is called the *transcendence degree* $\operatorname{trdeg}_K L$. Finally, show that, if $L = K(\alpha_1, \ldots, \alpha_n)$ any maximal algebraically independent subset of $\alpha_1, \ldots, \alpha_n$ is a transcendence basis. (In particular $\operatorname{trdeg}_K L \leq n$.)

Solution

We prove a result analogous to Proposition C.1.2: if $\alpha_1, \ldots, \alpha_m \in L$ are K-algebraically independent and $\beta_1, \ldots, \beta_n \in L$ are such that any element of L is algebraic over $K(\beta_1, \ldots, \beta_n)$, then $m \leq n$. Since transcendence bases satisfy both conditions, this shows that $\operatorname{trdeg}_K L$ is well-defined. This almost Exercise B.4.8†: any family of $n+1$ elements algebraic over $K(\beta_1, \ldots, \beta_n)$ is algebraically dependent over K. The only difference is that, in our case $\alpha_1, \ldots, \alpha_m$ are not necessarily in $K(\beta_1, \ldots, \beta_n)$. However, the first argument still works perfectly fine, the only difference is that, if α_i has degree d_i over $K(\beta_1, \ldots, \beta_n)$, we get (at most)

$$\prod_{i=1}^{m} d_i \frac{(mMN)^{n+1} - 1}{mMN - 1}$$

equations this time, which is still less than $(N+1)^m$ for large N if $m > n$.

For the second part, note that, by the same argument as Theorem 1.3.2 or by Chapter 6, any element of $K(\alpha_1, \ldots, \alpha_n)$ is algebraic over $K(S)$, where $S \subseteq \{\alpha_1, \ldots, \alpha_n\}$ is a maximal subset of K-algebrically independent element. ■

Exercise B.4.10†. Let K be an algebraically closed field which is contained in another field L. Suppose that $f_1, \ldots, f_m \in K[X_1, \ldots, X_n]$ are polynomials with a common root in L. Prove that they also have a common root in K.

Solution

We present two solutions, one based on Hilbert's Nullstellensatz from Exercise B.4.6† and one in characteristic 0 based on transcendence basis from Exercise B.4.9†. For the first sol, note that $f_1, \ldots, f_m$ have a common root in L if and only if there are no $g_1, \ldots, g_m \in L[X_1, \ldots, X_n]$ such that $f_1 g_1 + \ldots + f_m g_m = 1$. In that case, there are no such g_i in $K[X_1, \ldots, X_n]$ either, so $f_1, \ldots, f_m$ also have a common root in K.

We new present the second solution, which is perhaps more intuitive as it "lifts" (or "reduces" in our case) the common root over L to a common root over K. Thus, suppose that $\operatorname{char} K = 0$

and let $\alpha_1, \dots, \alpha_k$ be a K-transcendence basis for the field generated by K and the common root. Then, let α be such that this field is equal to $K(\alpha_1, \dots, \alpha_k, \alpha)$, there exists such an α by the primitive element theorem 6.2.1. Let

$$r_1(\alpha_1, \dots, \alpha_k)(\alpha), \dots, r_n(\alpha_1, \dots, \alpha_k)(\alpha))$$

be the common root, with $r_i \in K(X)$. The equality

$$f_i(r_1(\alpha_1, \dots, \alpha_k)(\alpha), \dots, r_n(\alpha_1, \dots, \alpha_k)(\alpha)) = 0$$

is an equality modulo the minimal polynomial $\pi(\alpha_1, \dots, \alpha_k)$ of α. Thus, if we replace α_i by $a_i \in K$ and α by a root $a \in K$ of $\pi(a_1, \dots, a_k)$, we get a common root in K. We just need to check that the $r_i(a_1, \dots, a_k)(a)$ are well-defined, i.e. their denominator is non-zero. This follows from Exercise A.1.7*: the denominator is non-zero so it stays non-zero infinitely many times in K^n. Note that $r_i(\alpha)$ is not necessarily a polynomial, instead it is algebraic over $K(\alpha_1, \dots, \alpha_k)$, but by considering its norm (the product with its conjugates over $K(\alpha_1, \dots, \alpha_k)$) we can get a polynomial. Indeed, if the norm of $r_i(\alpha)$ is non-zero then so is $r_i(\alpha)$. (We also need to be careful with the leading coefficient of π: if it vanishes α has too few conjugates and things can get weird, but we can simply pick $a_1, \dots, a_k$ so that it doesn't vanish either.) ■

Miscellaneous

Exercise B.4.11† (ISL 2020 Generalised)**.** Let $n \geq 1$ be an integer. Find the maximal N for which there exists a monomial f of degree N which can not be written as a sum

$$\sum_{i=1}^{n} e_i f_i$$

with $f_i \in \mathbb{Z}[X_1, \dots, X_n]$.

Solution

The answer is $N = \frac{n(n-1)}{2}$. First, we prove that $X_2 X_3^2 \cdot \ldots \cdot X_n^{n_1}$ can not be written in the desired form. Suppose for the sake of a contradiction that $X_2 X_3^2 \cdot \ldots \cdot X_n^{n_1} = \sum_i e_i f_i$ for some polynomials f_i, which we suppose without loss of generality to be homogeneous of degree $\frac{n(n-1)}{2} - i$ (by ignoring all other monomials). Then, we sum $\varepsilon(\sigma) X_{\sigma(2)}^1 \cdot \ldots \cdot X_{\sigma(n)}^{n-1}$ over all permutations $\sigma \in \mathfrak{S}_n$ of $[n]$, where ε denotes the signature (see Definition C.3.2). Since the e_i are symmetric, we have

$$\sum_{\sigma \in \mathfrak{S}_n} \varepsilon(\sigma) X_{\sigma(2)}^1 \cdot \ldots \cdot X_{\sigma(n)}^{n-1} = \sum_i e_i \sum_{\sigma \in \mathfrak{S}_n} \varepsilon(\sigma) f_i(X_{\sigma(1)}, \dots, X_{\sigma(n)}).$$

Here is the key point: if f has degree less than $\frac{n(n-1)}{2}$, $\sum_{\sigma \in \mathfrak{S}_n} \varepsilon(\sigma) f(X_{\sigma(1)}, \dots, X_{\sigma(n)}) = 0$. This is an obvious contradiction as the LHS is a sum of distinct monomials so is non-zero. To prove this claim, suppose without loss of generality that f is a monomial $\prod_{i=1}^n X_i^{a_i}$. Since $\sum_{i=1}^n a_i < \sum_{i=1}^n (i-1)$, two a_i must be equal, say $a_i = a_j$. Denote by τ the transposition $i \leftrightarrow j$. Then, by grouping permutations of $[n]$ by orbits $\sigma, \sigma \circ \tau$, the sum is zero since

$$f(X_{\sigma(1)}, \dots, X_{\sigma(n)}) = f(X_{\sigma \circ \tau(1)}, \dots, X_{\sigma \circ \tau(n)})$$

but $\varepsilon(\sigma \circ \tau) = -\varepsilon(\sigma)$ by Exercise C.3.11* so the sum over each orbits cancels out.

It remains to prove that $X_1^{a_1} \cdot \ldots \cdot X_n^{a_n}$ works when $a_1 + \ldots + a_n > \frac{n(n-1)}{2}$. When $a_1, \dots, a_n \geq 1$ it is trivial since the monomial is divisible by e_1. We proceed by induction on $a_1^2 + \ldots + a_n^2$, with the following base case: $a_1, \dots, a_n \geq 1$ the monomial is divisible by e_1.

Now suppose that $a_1 + \ldots + a_n > \frac{n(n-1)}{2}$ and, without loss of generality, $0 = a_1 \leq a_2 \leq \ldots \leq a_n$. There must exist some k such that $e_{k+1} \geq e_k + 2$, since otherwise $e_k \leq k-1$ for all k contradicting our initial assumption on the sum. Now consider

$$X_1^{a_1} \cdot \ldots \cdot X_n^{a_n} - X_1^{a_1} \cdot \ldots \cdot X_{k-1}^{a_{k-1}} X_k^{a_k - 1} \cdot \ldots \cdot X_n^{a_n - 1} e_{n-k}.$$

We claim that the sum of the squares of the exponents in any monomial appearing in this polynomial is less than $a_1^2 + \ldots + a_n^2$, thus concluding the inductive step. To see this, express a monomial of $X_1^{a_1} \cdot \ldots \cdot X_{k-1}^{a_{k-1}} X_k^{a_k - 1} \cdot \ldots \cdot X_n^{a_k} e_{n-k}$ as

$$X_1^{a_1 + b_1} \cdot \ldots \cdot X_{k-1}^{a_{k-1} + b_{k-1}} X_k^{a_k + b_k - 1} \cdot \ldots \cdot X_n^{a_n + b_n - 1}$$

for some $b_i \in \{0, 1\}$ with $b_1 + \ldots + b_n = n - k$. The wanted result then follows from the convexity of the square function: if $b_i = 1$ for some $i < k$ and $b_j = 0$ for some $j \geq k$, then $(a_i + 1)^2 + (a_j - 1)^2 < a_i^2 + a_j^2$. Iterating this process to "push" all the ones to the positions greater than or equal to k, we get

$$(a_1 + b_1)^2 + \ldots + (a_{k-1} + b_{k-1})^2 + (a_k + b_k - 1)^2 + \ldots + (a_n + b_n - 1)^2 \leq a_1^2 + \ldots + a_n^2$$

with equality if and only if we already had equality in the beginning, i.e. if the monomial is $X_1^{a_1} \cdot \ldots \cdot X_n^{a_n}$. However, we have ruled that case out by subtracting precisely this monomial, so we are done. ■

Exercise B.4.12† (Lagrange)**.** Given a rational function $f \in K[X_1, \ldots, X_n]$, we denote by G_f the set of permutations $\sigma \in \mathfrak{S}_n$ such that

$$f(X_1, \ldots, X_n) = f(X_{\sigma(1)}, \ldots, X_{\sigma(n)}).$$

Let $f, g \in K(X_1, \ldots, X_n)$ be two rational functions. If $G_f \subseteq G_g$, prove that there exists a rational function $r \in K[e_1, \ldots, e_n](X)$ such that

$$g = r \circ f.$$

Solution

We present the proofs in Prasolov [21]. Partition G_g into disjoint cosets $G_f = h_1 G_f, h_2 G_f, \ldots, h_k G_f$ and write $f_i = h_i f$ and $g_i = h_i f$ for each i, where σf means $f(X_{\sigma(1)}, \ldots, f(X_{\sigma(n)})$ (we say the group of permutations $\mathfrak{S}_n$ *acts* on the field $K(X_1, \ldots, X_n)$). (This is where we use the assumption that $G_f \subseteq G_g$.)

For the first proof, notice that

$$\sum_{i=1}^{k} \frac{g_i}{T - f_i}$$

is, by definition, symmetric in $X_1, \ldots, X_n$. Since $\Omega = \prod_{i=1}^{k} T - f_i$ is as well, we get

$$\sum_{i=1}^{k} \frac{g_i}{T - f_i} = \frac{F(T)}{\Omega(T)}$$

for some $F \in K(e_1, \ldots, e_n)[T]$ by the fundamental theorem of symmetric polynomials. Notice that $F'(f) = \prod_{i=2}^{k} f - f_i$ is $F/(T - f)$ evaluated at $T = f$ by Exercise 3.2.2*. Hence, we conclude that

$$\frac{F(f)}{\Omega'(f)} = \sum_{i=1}^{k} g_i \left(\frac{\Omega}{(T - f_i)\Omega'} \right)(f) = g$$

since $\frac{\Omega}{(T - f_i)\Omega'}$ vanishes at $f \neq f_i$.

The second proof is perhaps more intuitive. We consider the system of equations

$$\sum_{i=1}^{k} f_i^s g_i = T_s,$$

where the exponent represents powers and not iterates. Cramer's rule from Exercise C.5.7 and the Vandermonde determinant from Appendix C and tell us that

$$g = \frac{D}{\Delta}$$

where

$$\Delta = \begin{vmatrix} 1 & \cdots & 1 \\ f_1 & \cdots & f_n \\ \vdots & \ddots & \vdots \\ f_1^{k-1} & \cdots & f_k^{k-1} \end{vmatrix} = \prod_{i<j} f_i - f_j$$

and

$$D = \begin{vmatrix} T_0 & 1 & \cdots & 1 \\ T_1 & f_2 & \cdots & f_k \\ \vdots & \vdots & \ddots & \vdots \\ T_{k-1} & f_2^{k-1} & \cdots & f_k^{k-1} \end{vmatrix}.$$

Write this as $g = \frac{D\Delta}{\Delta^2}$. Notice that Δ^2 is symmetric, while D and Δ both change sign when two f_i are switched, so $D\Delta$ is symmetric in $f_2, \ldots, f_k$. However, it is easy to see that, for any i, $e_i(f_2, \ldots, f_k)$ can be expressed polynomially in terms of f_1 and $e_j(f_1, \ldots, f_k)$. Hence, this $\frac{D\Delta}{\Delta^2}$ is a rational function in f with symmetric coefficients by the fundamental theorem of symmetric polynomials. ■

Exercise B.4.13† (Iran Mathematical Olympiad 2012)**.** Prove that there exists a polynomial $f \in \mathbb{R}[X_0, \ldots, X_{n-1}]$ such that, for all $a_0, \ldots, a_{n-1} \in \mathbb{R}$,

$$f(a_0, \ldots, a_{n-1}) \geq 0$$

is equivalent to the polynomial $X^n + a_{n-1}X^{n-1} + \ldots + a_0$ having only real roots, if and only if $n \in \{1, 2, 3\}$.

Solution

If $n \leq 3$, the discriminant satisfies the condition. Indeed, the discriminant of $f = \prod_i X - \alpha_i$ is the square of $\prod_{i<j} \alpha_i - \alpha_j$ so is positive if all α_i are positive. It remains to prove that, for these n, $\prod_{i<j} \alpha_i - \alpha_j$ is real if and only if all α_i are (its square is real so it must be real or purely imaginary). For $n = 1$, it is trivial since any polynomial of degree 1 with real coefficients splits in $\mathbb{R}$. For $n = 2$, if the roots of f are $\alpha \neq \overline{\alpha}$, then $\alpha - \overline{\alpha}$ is not real since complex conjugation negates it. For $n = 3$, if the roots of f are $\alpha \neq \overline{\alpha}$ and $\beta \in \mathbb{R}$, then complex conjugation also negates

$$(\alpha - \overline{\alpha})(\beta - \alpha)(\beta - \overline{\alpha})$$

so it isn't real as desired.

Now, if there exists such a polynomial for $n \geq 4$, there exists one for $n = 4$ by setting $g(a, b, c, d) = f(a, b, c, d, 0, \ldots, 0)$. Thus, it only remains to prove that there doesn't exist such a polynomial for $n = 4$. For this, consider the special polynomial $f(0, b, 0, d)$ since we know precisely when the roots of $X^4 + bX^2 + d$ are real. For convenience, we shall in fact consider the polynomial $g(r, s) = f(0, -r - s, 0, rs)$ which is non-negative iff the roots of

$$X^4 - (r + s)X^2 + rs = (X^2 - r)(X^2 - s)$$

are all real, In other words, $g(r, s)$ is non-negative if and only if r and s are. This implies that

$$0 \geq \lim_{s \to 0^-} g(r, s) = g(r, 0) = \lim_{s \to 0^+} g(r, s) \geq 0,$$

i.e. $g(r, 0) = 0$ for any non-negative r. But then, the polynomial $g(R, 0)$ must be zero since it has infinitely many roots, so $g(r, 0)$ is also zero (and in particular non-negative) for negative r which is a contradiction. ■